

Saad Labyad

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Education

University of Oxford

Oxford, United Kingdom

PhD, Mathematics. (Grade: Awarded - Oxford PhDs are Pass/Fail only)

Oct 2018 - Oct 2024

Thesis: “Statistical learning of Hawkes models and market microstructure”.

Supervised by Prof. Samuel Cohen & Prof. Alvaro Cartea. Examined by Prof. Mathieu Rosenbaum & Prof. Mihai Cucuringu.

Funded by the Ioan and Rosemary James Scholarship and the Alan Turing Institute, affiliated with the Oxford-Man Institute.

- Developed scalable stochastic optimization methods for non-stationary Hawkes models, implemented the package `aslsd`.

- Conducted cross-sectional analysis of Nasdaq TotalView-ITCH data for high-frequency market impact modelling.

Other activities: Peer reviewer for Quantitative Finance (Taylor & Francis), co-supervisor of an MSc thesis (Hawkes models).

Selected coursework: Algorithmic Foundations of Learning, Rough Paths and Data Science, Theories of Deep Learning.

University of Oxford

Oxford, United Kingdom

MSc, Mathematical and Computational Finance. (Grade: Distinction – rank top 3)

Sep 2017 - Jul 2018

Thesis: “Topological persistence in market microstructure”, supervised by Prof. Samuel Cohen.

Selected coursework: Time Series, Machine Learning, Stochastic Calculus, Algorithmic Trading, Optimization, Asset pricing.

Ecole Centrale Paris

Paris, France

MEng, Applied Mathematics. (Grade: 4.11/4.33 GPA – rank top 3%)

Sep 2015 - Jul 2017

Projects: “Big data algorithms for NBO” (BEng); “Recommender systems for social networks” (MEng).

Selected coursework: Probability, Statistics, PDE, Algorithms, Computability, Risk Management, Microeconomics, Finance.

Lycée Masséna

Nice, France

Classes Préparatoires, Mathematics and Physics. (Grade: 176th/3370 national rank)

Sep 2012 - Jul 2015

Undergraduate course preparing students for the national competitive entrance exams to the French Grandes Ecoles.

Selected coursework: Linear Algebra, Analysis, Multivariate Calculus, Signal Processing, Topology, Quantum Physics.

Experience

Eleni

Cambridge, United Kingdom

Quantitative Researcher (Consultant)

Nov 2024 - Apr 2026

- Took time for family responsibilities, while conducting remote quantitative research work with Eleni (Cambridge Fintech).
- Developed and backtested cashflow forecasting pipelines on proprietary banking transactions data, reducing RMSE by 60%.

University of Oxford

Oxford, United Kingdom

Teaching Assistant

Oct 2018 - Oct 2024

- Taught MSc classes in Algorithmic Trading, Market Making, Stochastic Control, Stochastic Calculus, Exotic Derivatives.
- Conducted final mathematics interviews for applicants to the MSc in Mathematical Finance and the BA in Mathematics.

The Alan Turing Institute

London, United Kingdom

Doctoral Researcher

Mar 2022 - Feb 2023

- Developed clustering methods for market participants in collaboration with the Authority of Financial Markets (Netherlands).
- Analysed large-scale proprietary Euronext equities limit order book data detailing the market participants behind each order.

Ecole Centrale Paris

Paris, France

Research Assistant

Nov 2016 - Aug 2017

- Characterized closed-form equations for neo-classical utility functions from elasticities of substitution (Prof. Mehdi Senouci).
- Modelled bid-ask spreads in proprietary broker data using LSTM neural networks in TensorFlow (Prof. Ioane Muni-Toke).

Skills

Methods: Market Microstructure, Statistics, Optimization, Time Series, Point Processes, Machine Learning, Optimal Control.

Programming: Python (advanced – pandas, NumPy, SciPy, scikit-learn, PyTorch), C++ (intermediate), R (intermediate).

Languages: English (Fluent), French (Mother Tongue), Arabic (Mother Tongue), Spanish (Working Proficiency).

Selected Publications

- Labyad (2024). *Statistical learning of Hawkes models and market microstructure*. Oxford Thesis Collection.
- Cartea et al. (2024). *Statistical predictions of trading strategies in electronic markets*. Journal of Financial Econometrics
- Cartea et al. (2021). *Gradient-based estimation of linear Hawkes Processes with general kernels*. Preprint.
- Labyad & Senouci. (2018). *Deriving multiple-input production and utility functions from elasticities of substitution*. Preprint.

Activities

Label signed writer-composer, collaboration with major international music producers, sound engineering.